

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457567.8797 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.042
Transit depth (Rp/Rs)^2	1.98 +/- 1.17 [%]
Orbital Inclination (inc)	88.52 +/- 2.89
Airmass coefficient 1 (a1)	3.69 +/- 2.43
Airmass coefficient 2 (a2)	-1.04 +/- 0.66
Scatter in the residuals of the lightcurve fit is	74.46 %
Best Comparison Star	None
Optimal Aperture	4.47
Optimal Annulus	8.19
Transit Duration (day)	0.1301 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.042 vs 0.10538 (deviation 0.69 σ)
Duration fit vs published	0.1301 d vs 0.1167 d (deviation 1.29 σ)
Mid-time O-C	-5.5 min
Depth SNR	2.7
Residual scatter	74.46 %

Light curve

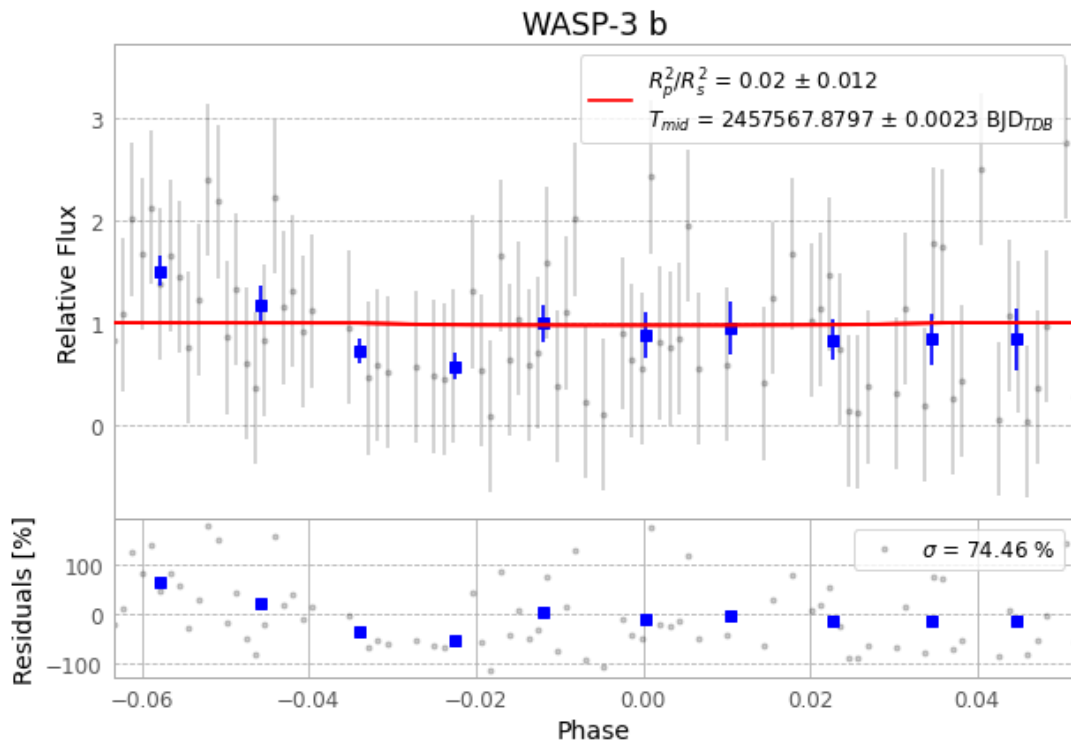


Figure 1 — FinalLightCurve_WASP-3 b_2016-06-27.png (SHA-256 ddb1686973338205...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [543, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5a289a561f4a0c58...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), WASP-3160628061212.FITS → WASP-3160628111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-160628.log (181cab5a8e71...), FinalLightCurve_WASP-3 b_2016-06-27.png (ddb168697333...), FinalLightCurve_WASP-3 b_2016-06-27.csv (b39b0426320d...)

Compute resources

Wall time 245.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 13:17Z.